

Climate Change Modeling Using IoT Sensor Data with Secure Analytics

Specific Use Case: Verifiable Climate Risk Intelligence as a Public Subscription Service

MS of Sciences in Professional Studies: Cybersecurity, Internet of Things, and Data Management & Analytics

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Problem Statement

The Climate Risk Intelligence Platform is a full end-to-end Internet of Things (IoT) system designed to collect environmental sensor data, compute real-time climate risk scores, and deliver transparent, verifiable insights through a modern, animated web portal.

PROBLEM

Current climate intelligence systems exhibit several limitations:

- Climate risk systems lack transparency
- IoT data pipelines are not verifiable
- Public access to risk intelligence is limited

SOLUTION

- Schema-controlled communication using Protocol Buffers,
- Distributed event-driven processing using MassTransit,
- Climate risk and confidence scoring mechanisms(flood, heat, air quality).
- Model evolution tracking and drift detection,
- Demonstrate a scalable model for public subscription-based climate intelligence

Challenges

- Hardware failures (BME280, ESP32 instability)
- Resolved via virtual sensor + hardware replacement

RESULTS

- Real-time IoT data successfully ingested and stored
- Protobuf pipeline verified (binary → decoded → database)
- R analytics engine executed:
 - Rolling averages Z-score
 - Anomaly detection
 - ARIMA forecasting
- Forecasts stored in database
- Frontend dashboard supports live + mock mode

Challenges

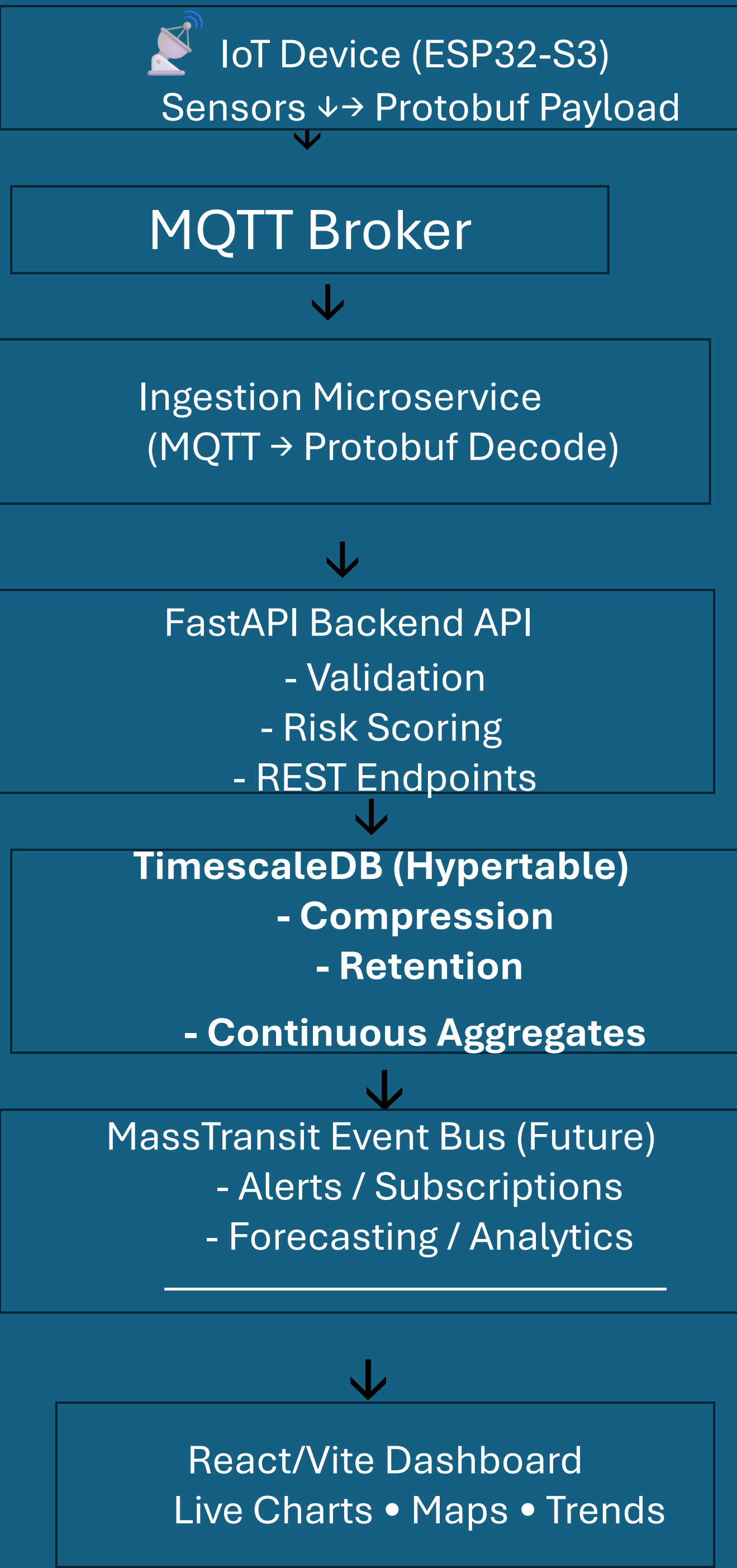
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Why It Matters

Communities need fast, trustworthy climate intelligence. This platform turns raw sensor data into actionable insights — enabling early warnings, environmental monitoring, and data-driven decision-making.

It shows transparency, not weakness.

SYSTEM ARCHITECTURE



Prototype system operational; final validation and extended testing in progress.

KEY FEATURES

- Deploy LoRaWAN gateway outdoors for long-range testing.
- Add ML-based flood prediction (Random Forest / LSTM).
- Add subscriber alerting (push notifications, webhooks).
- Expand sensor suite (soil moisture, PM2.5, river level).
- Deploy backend to cloud (Azure / AWS).
- Add user accounts + API keys for researchers

LESSONS

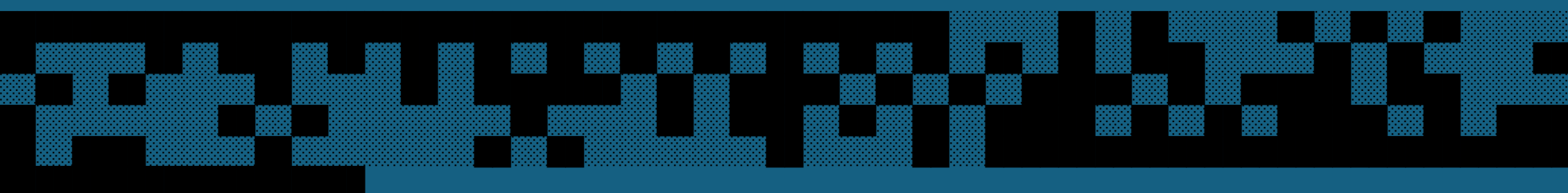
- Hardware reliability can impact system progress → simulation is essential
- Protobuf significantly improves performance for IoT systems
- End-to-end validation is critical before scaling architecture
- Modular design simplifies debugging and expansion
- Security must be integrated from the beginning

Acknowledgments

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QR CODE — Local Demo

Live Demo: URL: <http://localhost:5173>



QR CODE — GitHub Repository:

URL: <https://github.com/luciependesi>

